

Intention estimation

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1. Course Content

Basic: Master the ability to customize unique user intent understanding functionality using the RAG knowledge base.

Advanced: Master the debugging methods for intent understanding effects on the Dify platform.

[!IMPORTANT]

- Intent understanding is designed to increase the synergy between the robot and the human, allowing the robot to understand the user more uniquely. This function should not be used to perform "strange" and "illogical" tasks.

2. Start the Agent

Note: If it's already started, you don't need to start it again.

Enter the command on the vehicle terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:

```
mircoROS_Agent
[1767693398.286449] info | ProxyClient.cpp | create_datareader | d
subscriber created | client_key: 0x1D229138, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1767693398.290468] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x1D229138, topic_id: 0x007(2), participant_
id: 0x000(1)
[1767693398.294835] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x1D229138, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1767693398.299603] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x1D229138, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
[1767693398.303485] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x1D229138, topic_id: 0x008(2), participant_
id: 0x000(1)
[1767693398.330727] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x1D229138, subscriber_id: 0x003(4), partici
pant_id: 0x000(1)
[1767693398.335405] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x1D229138, datareader_id: 0x003(6), subscri
ber_id: 0x003(4)
```

2.1 Configure the Intent Mapping File

- This file is used to store your vague intents and the corresponding task operations you want the robot to perform.
- Open the example file in this tutorial folder. Following the reference format, you can add multiple customized intents. Here's a simple example:

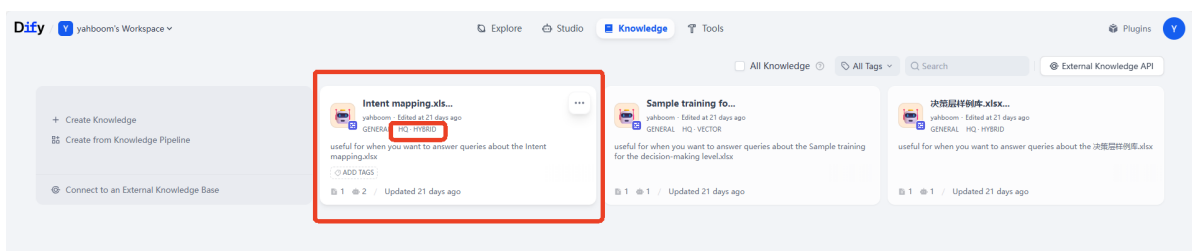
query	answer
I'm thirsty	1. Navigate to the kitchen, 2. Check if there's bottled water or a drink, 3. If a drink is available, return to the starting location and inform the user.

2.2 Configuring the Knowledge Base

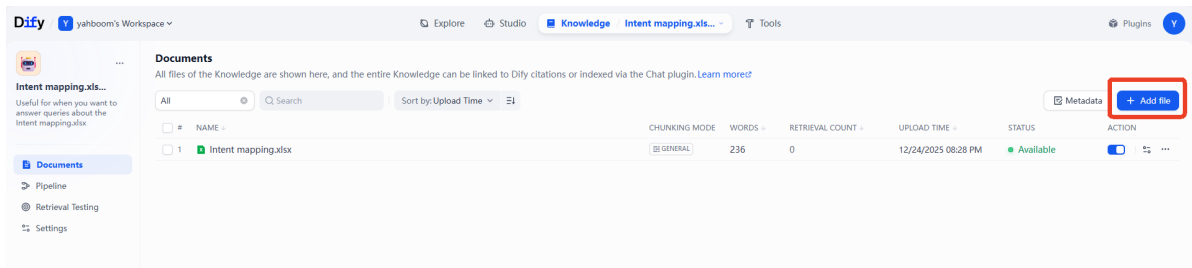
- Next, we need to upload our edited intent mapping file to the **RAG** knowledge base in **Diffy**.

[!TIP]

- For detailed instructions on operating the RAG knowledge base, please refer to the tutorial in **[04.AI Model Development]–[06.Deploy the RAG knowledge base]**.
- Diffy comes with a default **Intent mapping** reference knowledge base to demonstrate how to use intent understanding features.



- You can modify the **Intent mapping.xlsx** template, or you can delete it and add your own file.



[!TIP]

Note:

- To ensure optimal intent understanding performance, it is recommended to set the `Intent mapping` intent understanding knowledge base to high-quality mode, as intent understanding often requires retrieving relevant fragments between similar semantic cues.

3. Running Examples

3.1 Starting the Program

- On the vehicle's infotainment system, open the terminal and enter the command to start the AI agent function, or use the shortcut `multi_brains`:

```
ros2 launch m3_bringup llm_agent_control.launch.py
```

- Start the navigation function on the vehicle's infotainment system:

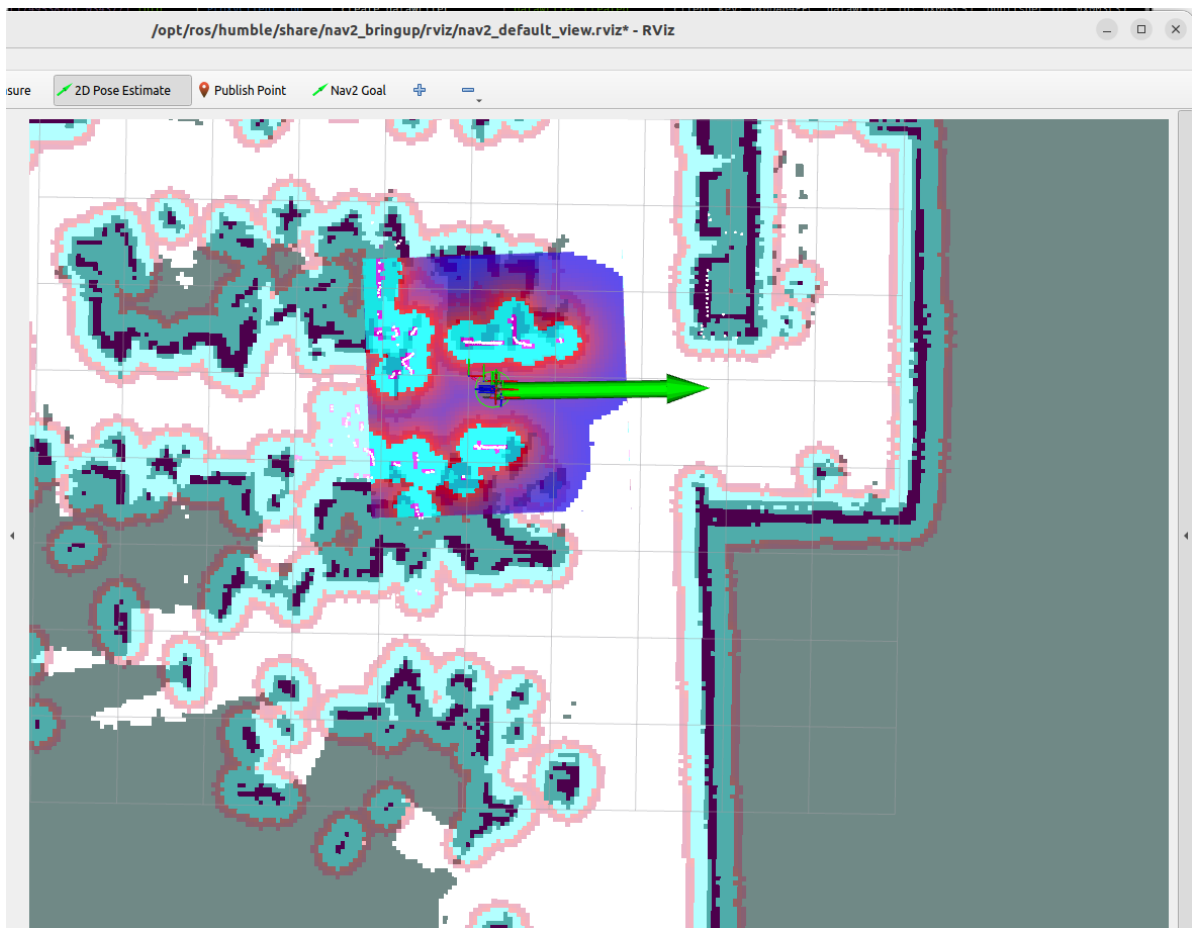
(ORIN & PI5 mainboard)

```
ros2 launch m3_bringup nav2.launch.py
```

(RDK mainboard) Run step-by-step, visualize the startup on the accompanying virtual machine `Rosmaster-M3-Ubuntu22.04`:

```
#Virtual machine side
rviz2 -d /home/yahboom/yahboomM3_ws/user_ws/src/m3_bringup/rviz/nav2.rviz
#Host side
ros2 launch m3_bringup nav2.launch.py gui:=False
```

Afterwards, follow the navigation startup process to initialize positioning. This will open the rviz2 visualization interface. Click on **2D Pose Estimate** in the top toolbar to enter selection mode. Roughly mark the robot's location and orientation on the map. After initializing positioning, the preparation is complete.



3.2 Test Cases

The test cases here are for reference only. Users can create their own dialogue commands.

- I'm currently in the master bedroom, and I'm feeling a bit thirsty.

```
asr-5] [INFO] [1750491006.444004139] [asr]: 0
asr-5] [INFO] [1750491006.534965377] [asr]: 0
asr-5] [INFO] [1750491006.624816840] [asr]: 0
asr-5] [INFO] [1750491006.715541268] [asr]: 0
asr-5] [INFO] [1750491006.805879567] [asr]: 0
asr-5] [INFO] [1750491006.897671209] [asr]: 0
asr-5] [INFO] [1750491006.989015568] [asr]: 0
asr-5] [INFO] [1750491007.079612532] [asr]: 0
asr-5] [INFO] [1750491007.171301831] [asr]: 0
asr-5] [INFO] [1750491007.263230564] [asr]: 0
asr-5] [INFO] [1750491007.354806162] [asr]: 0
asr-5] [INFO] [1750491007.445503515] [asr]: 0
asr-5] [INFO] [1750491007.536006330] [asr]: 0
asr-5] serial /dev/ttyUSB1 open
asr-5] funasr version: 1.2.6.
'load_data': '0.030', 'extract_feat': '0.021', 'forward': '1.870', 'batch_size': '1', 'rtf': '0.405',
tf_avg: 0.405: 100%|██████████| 1/1 [00:01<00:00, 1.87s/it]
tf_avg: 0.405: 100%|██████████| 1/1 [00:01<00:00, 1.87s/it]
asr-5] [INFO] [1750491009.522065243] [asr]: 我现在在主卧室我感觉我有点口渴了
asr-5] [INFO] [1750491009.522748441] [asr]: okay😊, let me think for a moment...
model_service-3] [INFO] [1750491011.870241207] [model_service]: The upcoming task to be carried out: 1.
导航前往厨房
model_service-3] 2. 观察夹取矿泉水
model_service-3] 3. 导航返回主卧室
model_service-3] 4. 放下矿泉水
```

The steps for planning the decision-level model are as follows:

```
model_service-3] [INFO] [1750491011.870241207] [model_service]: The upcoming task to be carried out: 1.
导航前往厨房
model_service-3] 2. 观察夹取矿泉水
model_service-3] 3. 导航返回主卧室
model_service-3] 4. 放下矿泉水
```

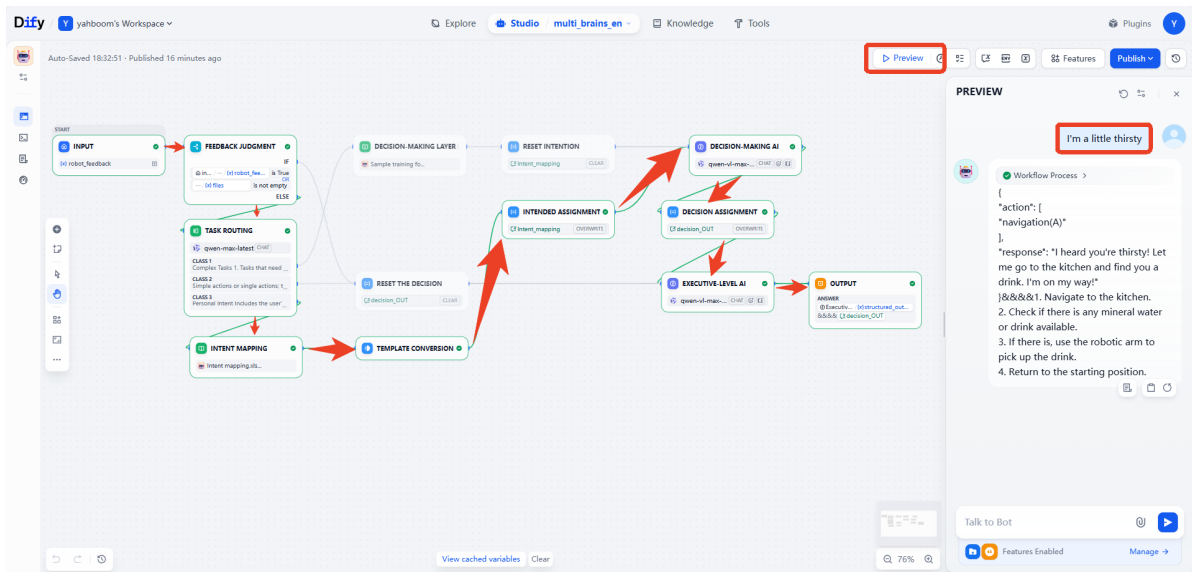
Following the steps for planning the large decision-level model, execute the following sequentially:

```
[model_service-3] [INFO] [1750491974.359941765] [model_service]: The upcoming task to be carried out: 导航前往厨房区域, 观察夹取矿泉水, 导航返回主卧室, 放下矿泉水。
[model_service-3] [INFO] [1750491977.376557382] [model_service]: "action": ['navigation(G)'], "response": "好的, 我这就去厨房帮你拿矿泉水。就像一个勇敢的骑士一样, 出发!"
[action_service-4] [INFO] [1750491985.052591699] [action_service]: navigation Goal accepted
[action_service-4] [INFO] [1750492007.324887601] [action_service]: Navigation succeeded!
[model_service-3] [INFO] [1750492010.560577885] [model_service]: "action": ['seewhat()'], "response": "我已经到达厨房了, 现在开始寻找矿泉水。让我用我的火眼金睛看看它在哪里!"
[model_service-3] [INFO] [1750492023.278365389] [model_service]: "action": ['grasp obj(305, 48, 392, 276)'], "response": "找到了! 矿泉水就在那里, 我这就去把它夹起来。看我的动作是不是很熟练?"
[kin_srv-2] [WARN] [1750492032.926281990] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a root link with an inertia. As a workaround, you can add an extra dummy link to your URDF.
[action_service-4] start
[action_service-4] [INFO] [1750492036.744005351] [image_converter]: Received: [305, 48, 392, 276]
```

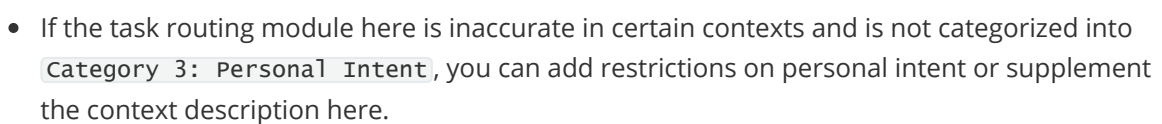
4. Effect Debugging

4.1 Visual Workflow

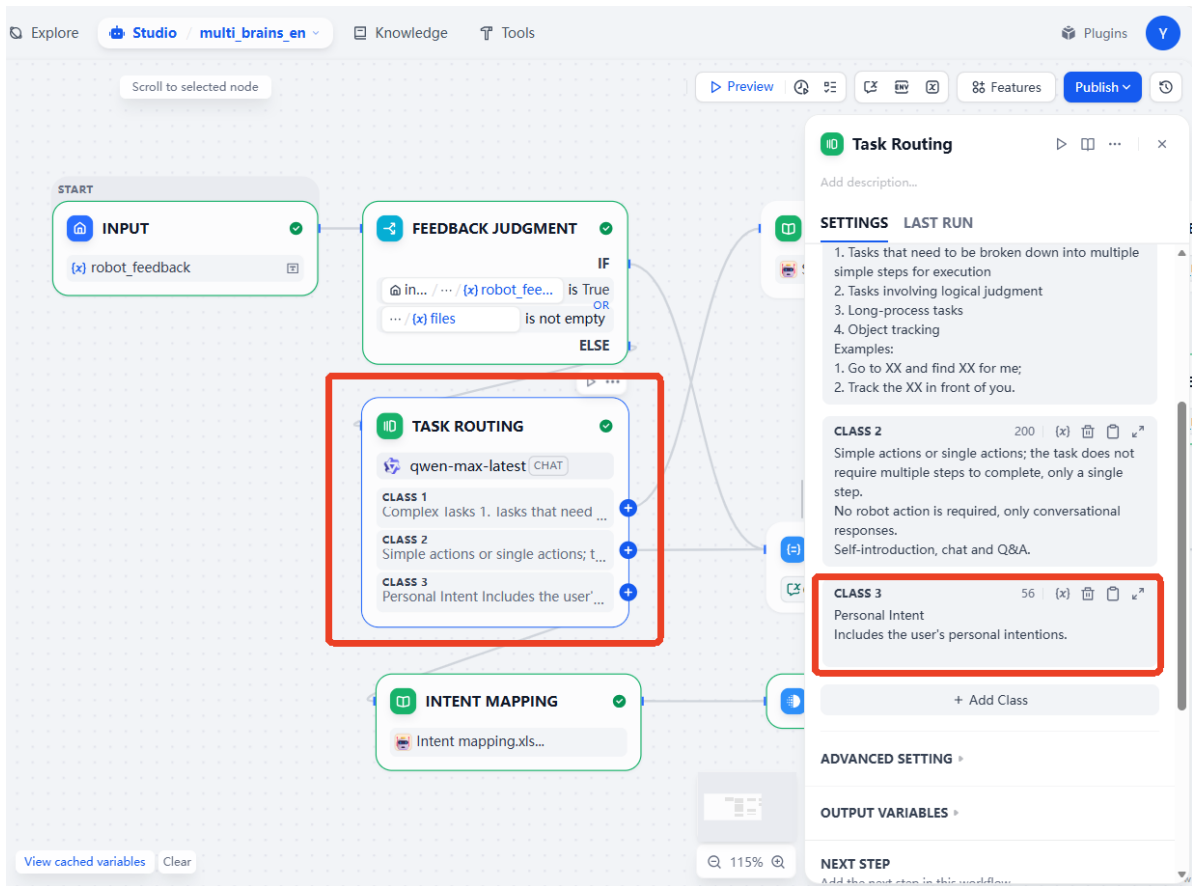
- Open the corresponding language version of the multi_brains application, then click preview. You can input test content to view the data flow.



- You can also open the workflow to view the time taken for each step and the input and output content of each process.



- If the task routing module here is inaccurate in certain contexts and is not categorized into `Category 3: Persona1 Intent`, you can add restrictions on personal intent or supplement the context description here.



- Additionally, the recall effect of the intent understanding statement to be tested in the knowledge base of the intent mapping can also be tested separately.

The screenshot shows the Dify Knowledge base interface with a 'Retrieval Test' result.

Retrieval Test:

- SOURCE TEXT:** I'm a little thirsty
- Test Button:** Test

Records:

QUERY CONTENT	SOURCE	TIME
I'm a little thirsty	Retrieval Test	01/14/2026 06:35 PM
I'm a little thirsty	App	01/14/2026 06:32 PM

1 Retrieved Chunks:

- Chunk-01:** 236 characters. **SCORE: 0.85**
- query:** "I'm a little thirsty"; **answer:** "1. Navigate to the kitchen. 2. Check if there is any mineral water or drink available. 3. If there is, use the robotic arm to pick up the drink. 4. The navigation returns to the..."
- Source:** Intent mapping.xls (OPEN)